

# CISC3003 Web Programming

## UM'S BICYCLE RENTAL MANAGEMENT SYSTEM

### TEAM 01

dc325182 – Che Chi Hin  
dc325420 – Chang I Wai  
dc325381 – Hoi Kai Cheng  
dc325107 – Ho Weng Hong  
dc325095 – Chung Hou Sam  
dc325837 – Lei Man Lam

Pair 07

Pair 09

Pair 14



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(BPMN,C4, UML)**

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# CONTEXT

## Background

The University of Macau (UM) campus is **large**, making **walking** between buildings **time-consuming** and resulting in **informal, disorganized** bicycle usage.

## Web Application Architecture

- **Frontend** → HTML, CSS, and JavaScript
- **Backend** → PHP, running on a local XAMPP server environment, InfinityFree (Free Web Hosting)
- **Database** → MySQL in InfinityFree





# GOALS

## Primary Objective

- provide a convenient "last-mile" transport option
- helps users move efficiently across the campus

## System Goals

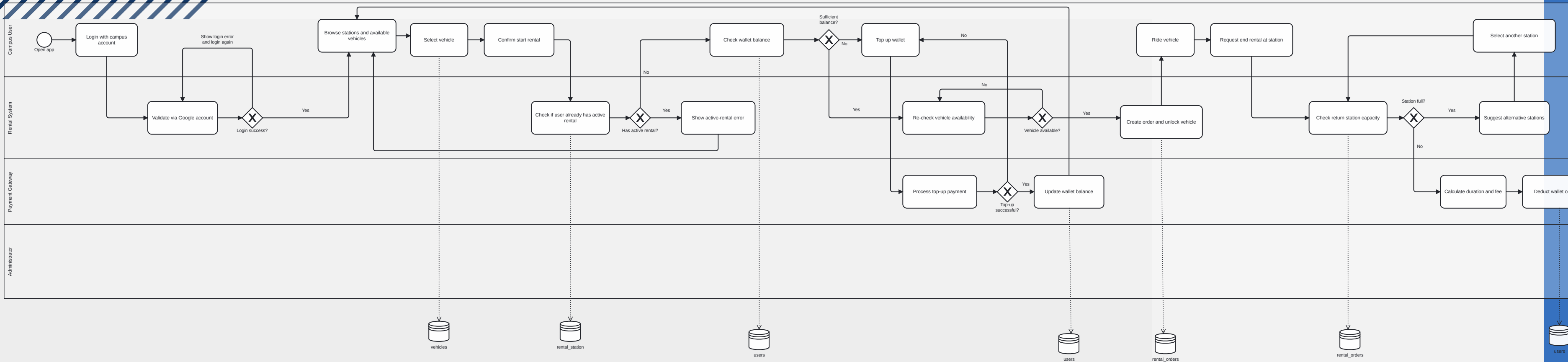
Design a platform (**register/log in/manage rental accounts**)

- **core operations** → searching, renting, returning, and viewing history.
- **administrators** → manage vehicles, stations, and accounts



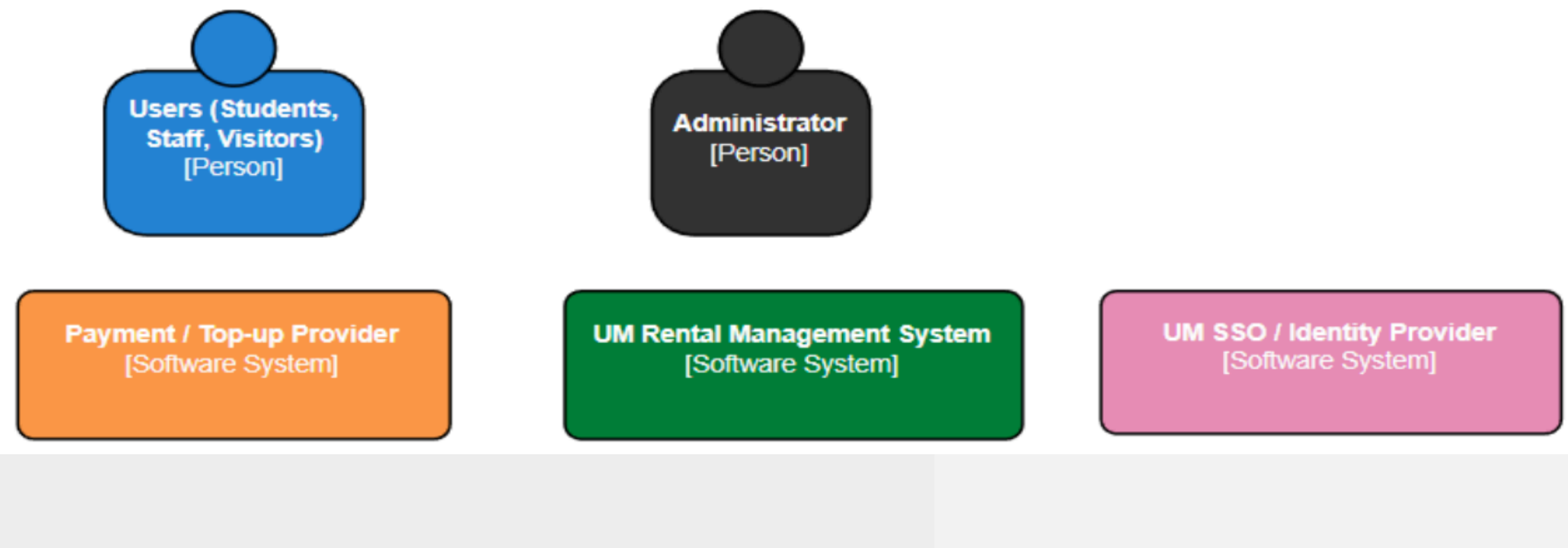
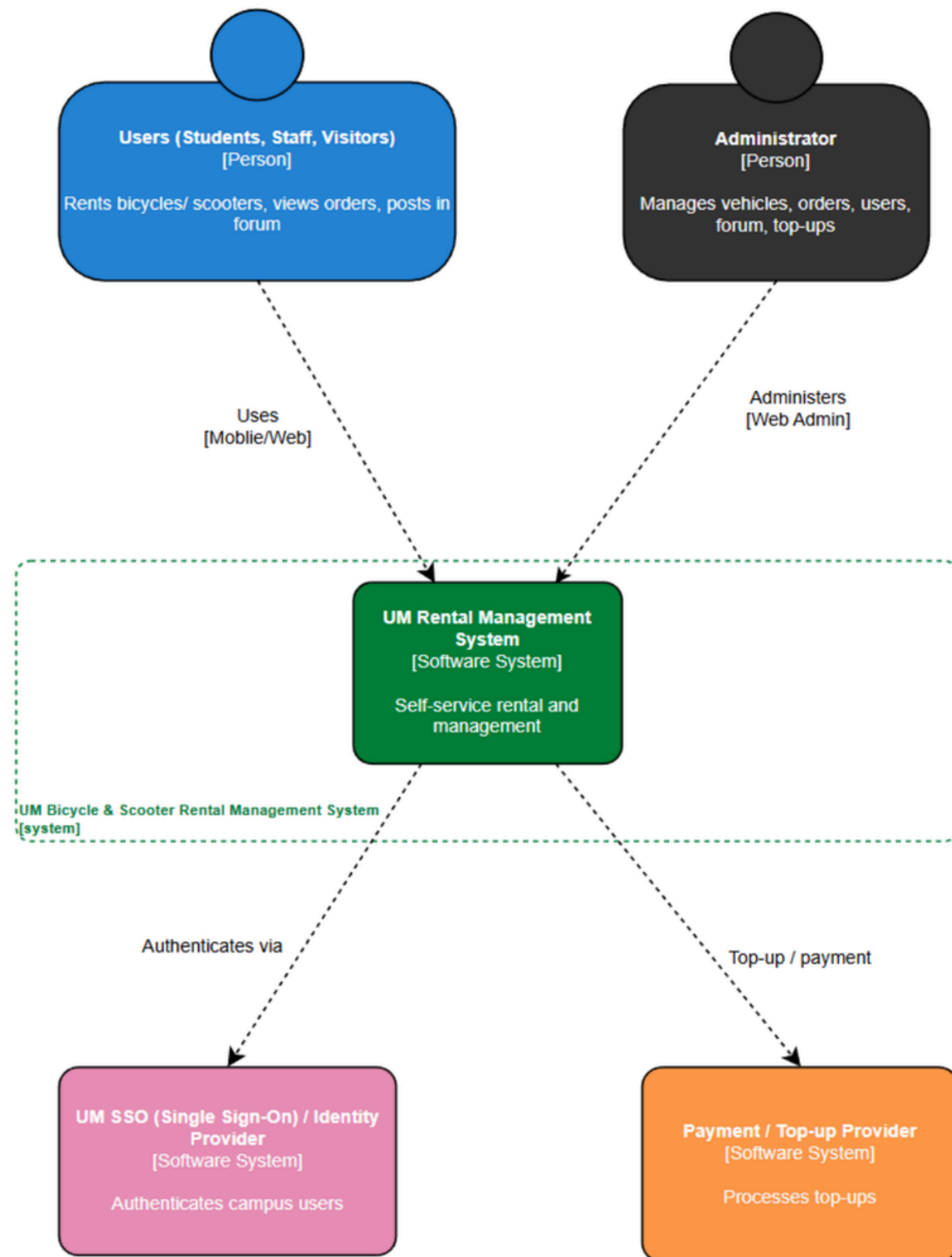


# BPMN



Speaker: dc325381 – Hoi Kai Cheng

# SYSTEM CONTEXT DIAGRAM

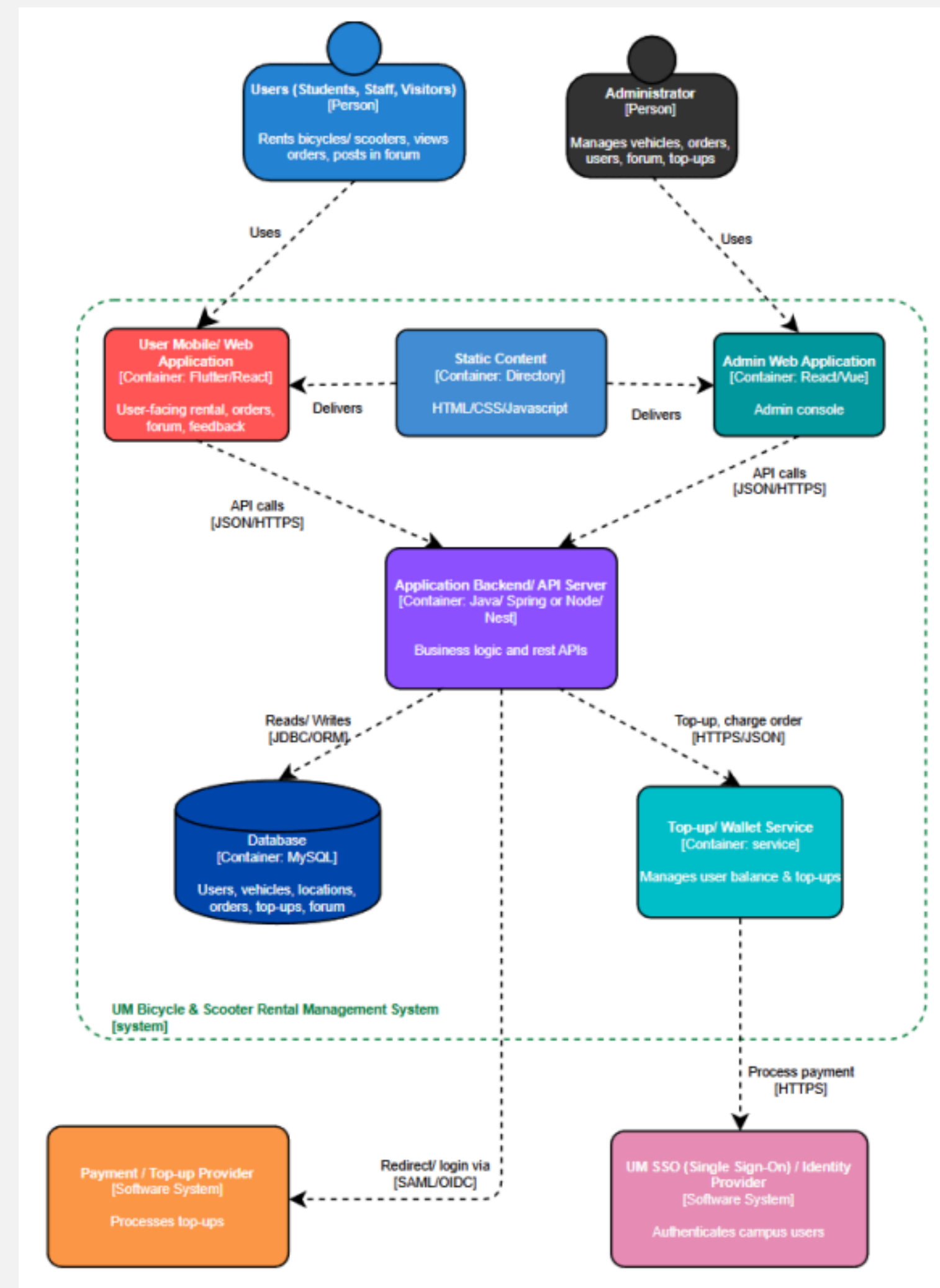


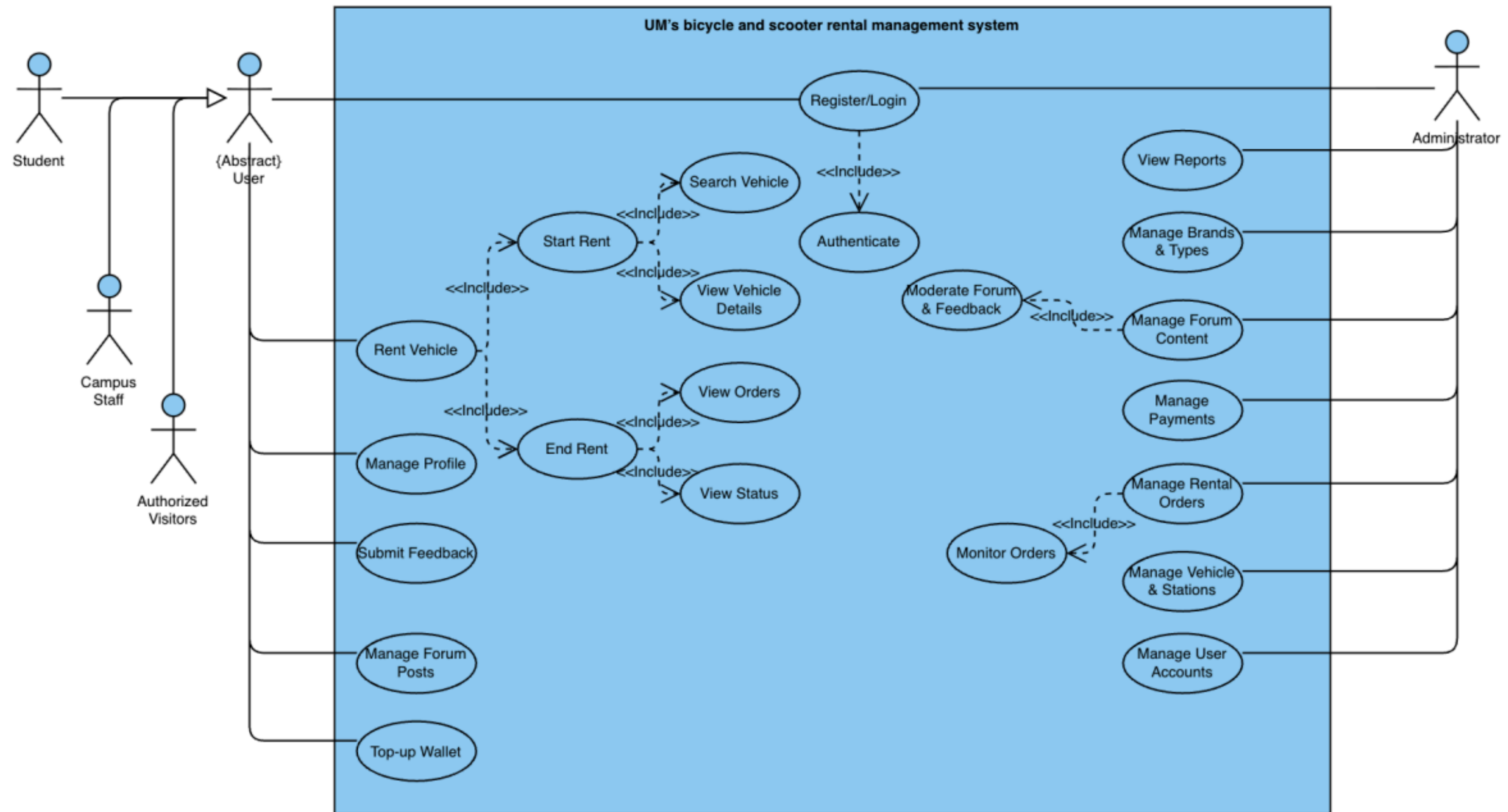
Speaker: dc325381 – Hoi Kai Cheng



- This diagram showcases the high-level architecture, showing how the PHP Backend connects the Frontend to the MySQL Database.

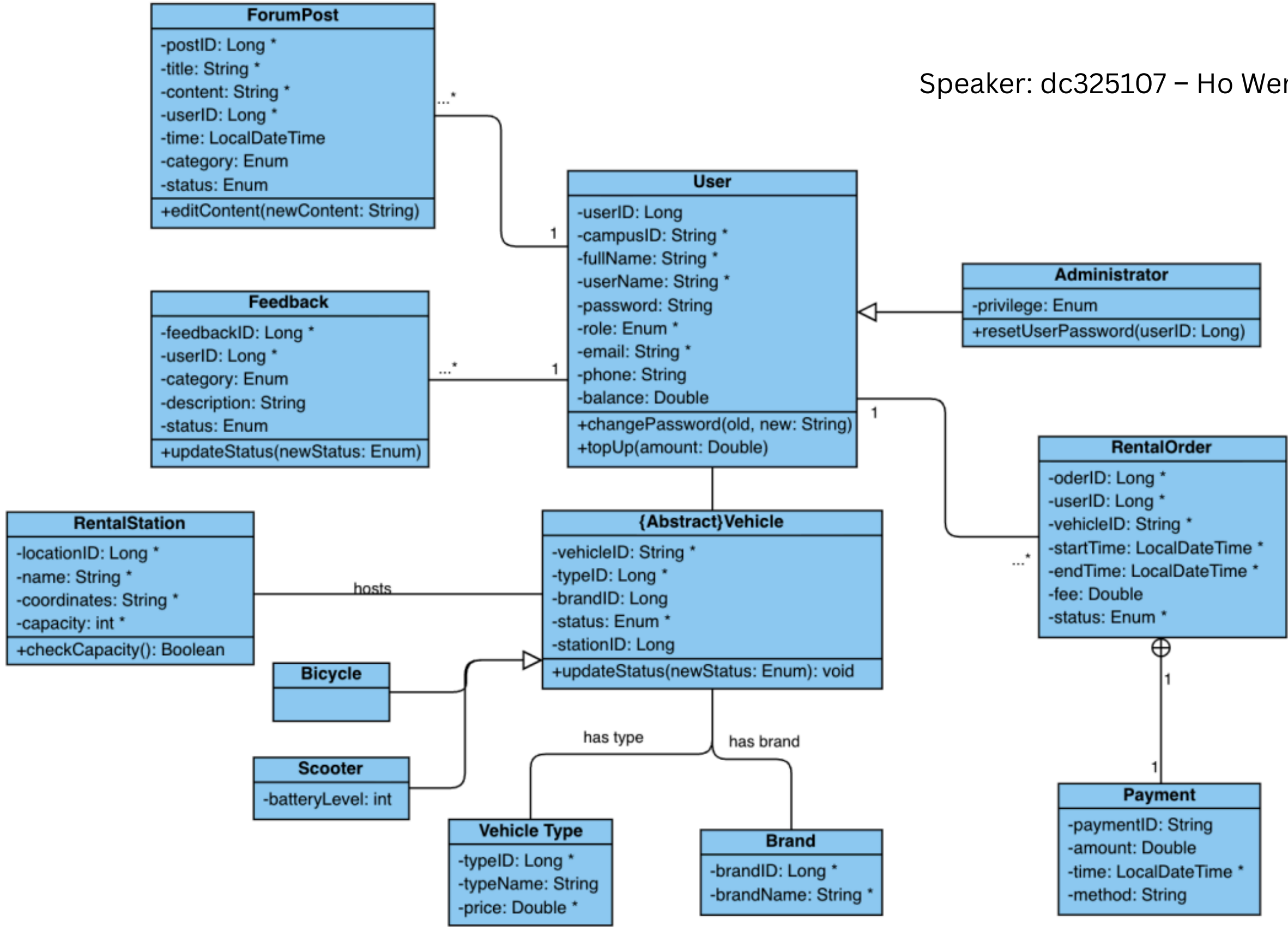
Speaker: dc325381 – Hoi Kai Cheng





Speaker: dc325107 – Ho Weng Hong

- This diagram maps out the full functional scope for both Campus Users and Administrators.



- This diagram defines the database entities and their relational attributes to ensure data normalization.



# SERVICES AND REQUIREMENTS

## Data Requirements (Entities)

### Users

ID, Role, Email, Balance.

### Vehicles & Stations

Vehicle Status, Type, Location  
Coordinates, Capacity Limits.

### Rental Orders

Start/End times, Fees, Status.

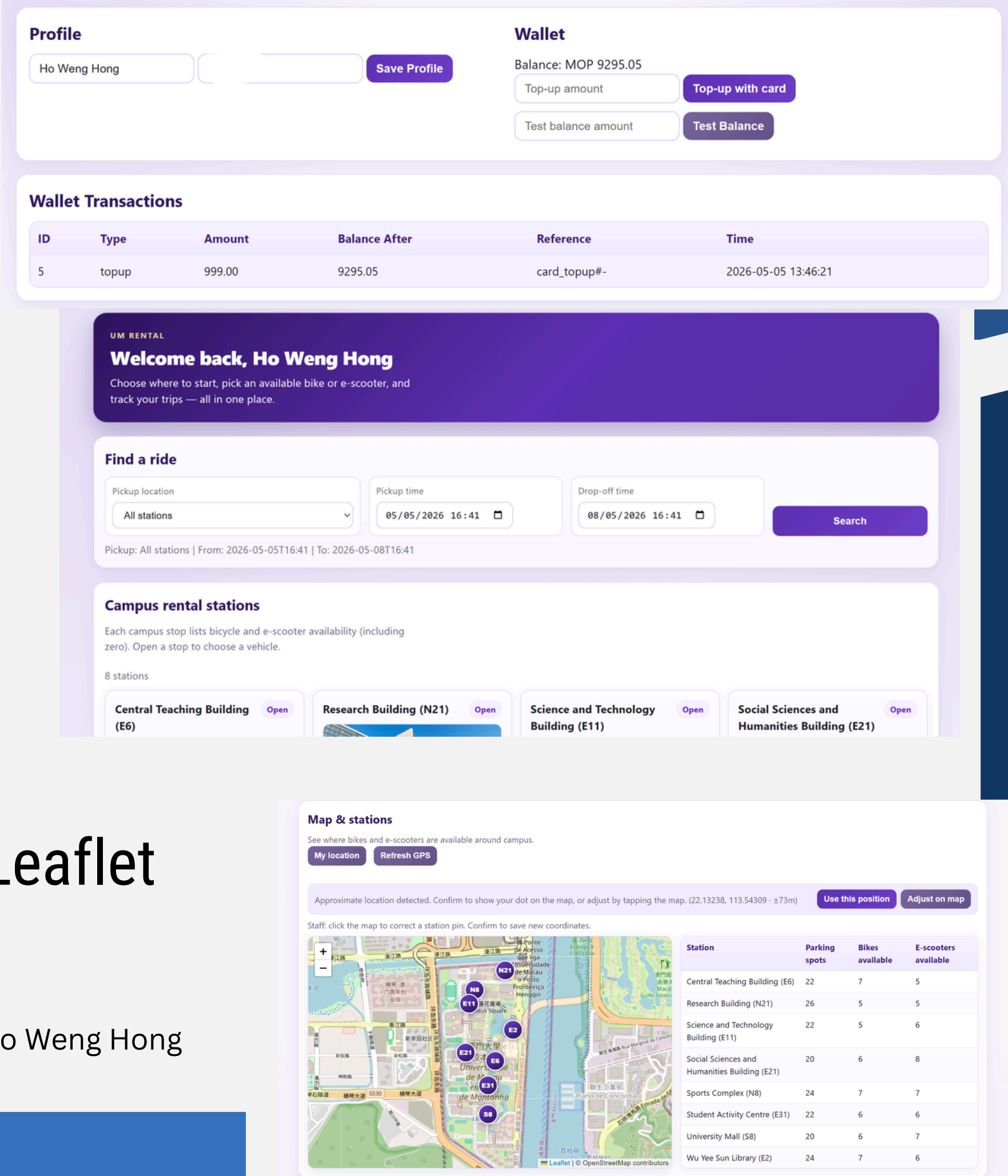
Speaker: dc325107 – Ho Weng Hong



# CORE SERVICES (USER)

- Account authentication via Google
- Profile management
- Vehicle search
- Click "Rent" on the rental website
- Wallet top-up
- Interactive campus map integrating Leaflet and OpenStreetMap

Speaker: dc325107 – Ho Weng Hong





# CORE SERVICES (ADMIN)

- Inventory management (add/retire vehicles)
- Station capacity control
- Manual order adjustments
- Forum moderation

Speaker: dc325107 – Ho Weng Hong

UM Rental

English

Home

Contact Us

My Account

User Dashboard

Staff

Logout

Admin portal

Admin Navigation

- Vehicle Management
- Staff Accounts
- Station Status
- Abnormal Orders
- Reports

Vehicle Dispatch & Status Update

| ID  | Model   | Brand   | Status    | Station                            | Action                                                                                                |
|-----|---------|---------|-----------|------------------------------------|-------------------------------------------------------------------------------------------------------|
| 120 | scooter | Ninebot | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 119 | scooter | Xiaomi  | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 118 | scooter | Giant   | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 117 | scooter | Ninebot | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 116 | scooter | Xiaomi  | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 115 | bicycle | Xiaomi  | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 114 | bicycle | Giant   | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 113 | bicycle | Ninebot | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 112 | bicycle | Xiaomi  | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 111 | bicycle | Giant   | available | University Mall (S8) (#8)          | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 110 | scooter | Ninebot | available | Student Activity Centre (E31) (#7) | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 109 | scooter | Xiaomi  | available | Student Activity Centre (E31) (#7) | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |
| 108 | scooter | Giant   | available | Student Activity Centre (E31) (#7) | <div>available (可租借) ▼</div> <div>Update Status</div> <div>Station ID</div> <div>Assign Station</div> |

UM Rental

English

Home

Contact Us

My Account

User Dashboard

Staff

Logout

Admin portal

Admin Navigation

- Vehicle Management
- Staff Accounts
- Station Status
- Abnormal Orders
- Reports

Reports

Total Orders

21

Total Revenue

MOP 3365.25

Avg Minutes

574

completed

19

payment\_pending

1

force\_closed

1

01/05/2026

05/05/2026

Load KPI Report

| Station                                       | Total Orders | Revenue |
|-----------------------------------------------|--------------|---------|
| Research Building (N21)                       | 5            | 1462.20 |
| Central Teaching Building (E6)                | 3            | 710.75  |
| Social Sciences and Humanities Building (E21) | 3            | 202.90  |
| Wu Yee Sun Library (E2)                       | 3            | 78.60   |
| Science and Technology Building (E11)         | 2            | 278.30  |
| Sports Complex (N8)                           | 2            | 5.50    |
| University Mall (S8)                          | 2            | 341.00  |
| Student Activity Centre (E31)                 | 1            | 286.00  |



# PROJECT TASKS



## PHASE 1

Project Planning and UI/UX Design, including finalizing the proposal and visual layouts.



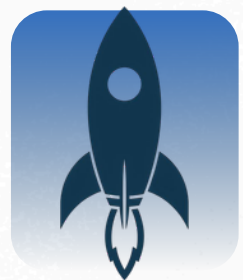
## PHASE 2

Front-End Development, translating designs into HTML/CSS/JS and building core rental interface workflows



## PHASE 3

Back-End Integration, configuring XAMPP/MySQL and writing PHP scripts for authentication, fee calculation, and real-time status updates



## PHASE 4

System Testing and Deployment via InfinityFree

Speaker: dc325182 – Che Chi Hin

# CHALLENGES

## STATE MANAGEMENT

- Accurately tracking vehicle statuses (**Available, Rented, Maintenance**)
- preventing multiple active rentals **simultaneously**

## CAPACITY CONSTRAINTS

### Implementing logic

- preventing ending a rental at a station
- and exceeded its maximum vehicle capacity

## SECURITY

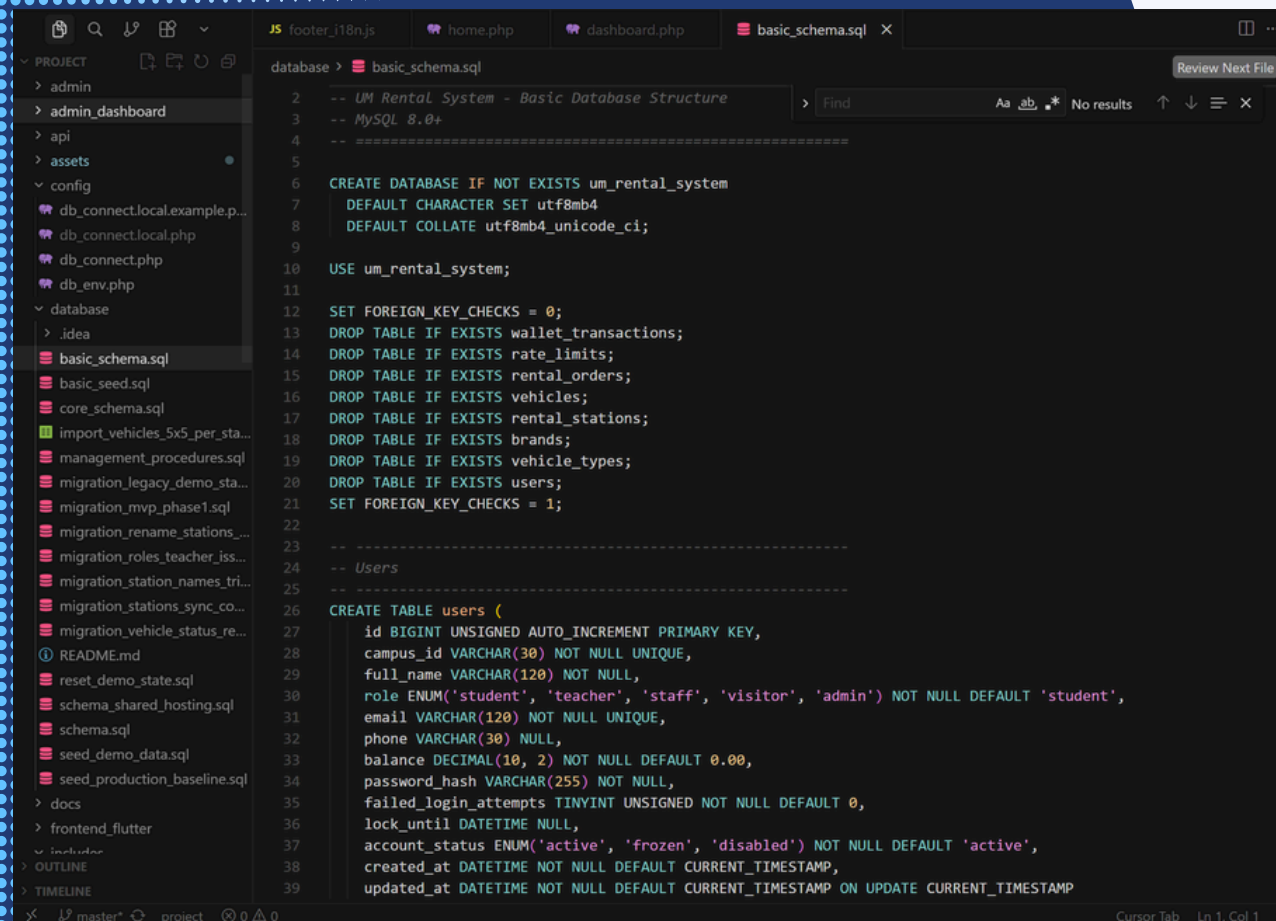
- password hashing
- role-based access control  
(users/ administrators)

Speaker: dc325420 – Chang I Wai

# DELIVERABLES

- A fully functional, full-stack web application hosted on InfinityFree.
- A complete Full website source code (HTML, CSS, Javascript, PHP)
- Database script, content & EER (Enhanced ER Diagram)
- UML/BPMN diagram

Speaker: dc325420 – Chang I Wai





# DIVISION OF WORK AMONG TEAM MEMBERS

## Pair 09

### **Hoi Kai Cheng & Ho Weng Hong**

Backend architecture, PHP scripting, and MySQL database design

## Pair 07

### **Che Chi Hin & Chang I Wai**

Frontend UI/UX design, responsive HTML/CSS layouts, and JavaScript interactions

## Pair 14

### **Chung Hou Sam & Lei Man Lam**

System testing, UML documentation, admin portal services, and GitHub version control management

Speaker: dc325095 – Chung Hou Sam



# CONTRIBUTIONS FROM TEAM MEMBERS



**Hoi Kai Cheng**

Team Leadership, Core Backend/Database  
Architecture, Deployment setup.



**Ho Weng Hong**

Staff & Admin Portal Logic, Role-based Access  
Control.



# CONTRIBUTIONS FROM TEAM MEMBERS



**Che Chi Hin**

Frontend UI/UX Redesign, Modal implementations.



**Chang I Wai**

Leaflet Map API Integration, Geolocation logic.



# CONTRIBUTIONS FROM TEAM MEMBERS



**Chung Hou Sam**

Database Schema design, Migrations, Seed Data.

Speaker: dc325095 – Chung Hou Sam



**Lei Man Lam**

i18n Multi-language Integration, Documentation and successfully deployed the final system platform



# CRITERIA OF SUCCESS

## Performance

System supports 50 concurrent users with  $< 2$  seconds response time for standard operations

## Reliability

Preserves committed database transactions (start/end rentals) even during unexpected failures

## Functionality

over lots of the core modules (Signup, Login, Dashboard, Rental Workflows) operate without critical bugs





# CONTRIBUTIONS FROM TEAM MEMBERS

**HO WENG HONG**

**Che Chi Hin**

**Hoi Kai Cheng**

**Chang I Wai**



**Chung Hou Sam**

**Lei Man Lam**

Speaker: dc325095 – Chung Hou Sam

# LIVE SYSTEM DEMO



<https://cisc3003team01.gt.tc>



# PROJECT EVALUATION

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## System Testing Validation

- Passing all predefined test cases
- (e.g., preventing rentals at full stations, successful timeout logouts)



## Instructor Assessment

- Successful demonstration of the deployed Web URL
- Improving the integration of the frontend and backend modules



Speaker: dc325837 – Lei Man Lam

# PROJECT LEARNING

## INDIVIDUAL LEARNING

Mastering specific technical stacks (HTML, CSS, PHP, SQL) and understanding software engineering principles like UML modeling and requirement specification

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## TEAM LEARNING

utilizing GitHub for collaborative version control, and conducting structured strategic meetings





# DECLARATION OF AI USAGE

AI Tool: Google Gemini.

Application: Documentation refinement, proposal structuring, and code logic troubleshooting.

Review: All content verified and adapted by the team to ensure academic integrity.



Speaker: dc325837 – Lei Man Lam

# REFERENCES (APA STYLE)

Chonoles, M. J., & Schardt, J. A. (2003). UML 2 for dummies. Wiley.

Mozilla. (2025). *MDN Web Docs: Responsive Web Design*.

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[US/docs/Learn\\_web\\_development/Core/CSS\\_layout/Responsive\\_Design](https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design)

Seidl, M., Scholz, M., Huemer, C., & Kappel, G. (2015). *UML @ Classroom: An introduction to object-oriented modeling*. Springer.





# THANK YOU

**Any Questions?**

